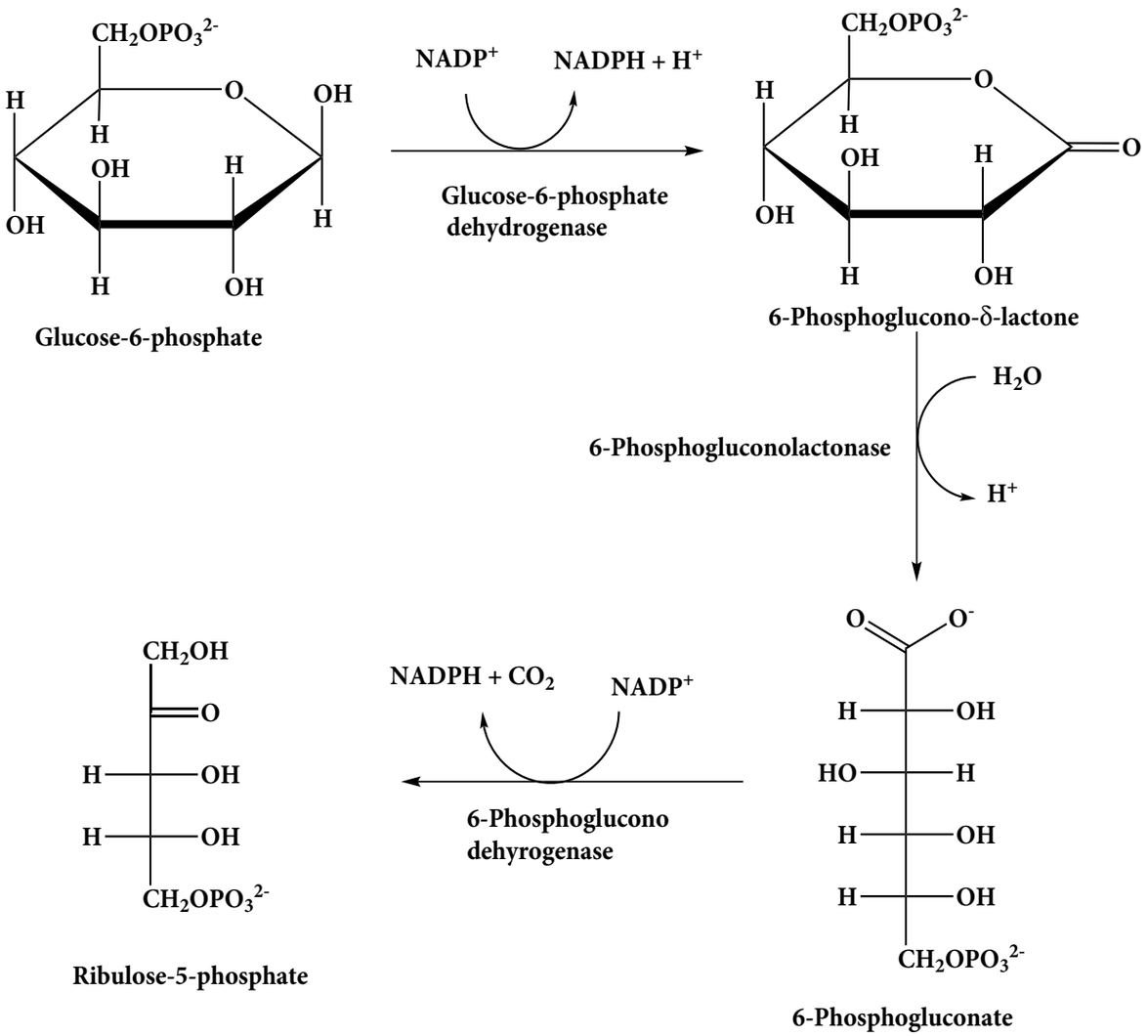
3.5.1. Reactions of oxidative phase

Three enzymes are required in the oxidative phase:

Step 1: In the first step, glucose-6-phosphate dehydrogenase oxidizes glucose-6-phosphate to 6-phosphoglucono-δ-lactone and reduces one molecule of $NADP^+$ to NADPH.

Step 2: In the second step, gluconolactonase cleaves the internal ester bond, which gives 6-phosphogluconate.

Step 3: In the third step, 6-Phosphogluconate dehydrogenase reduces another molecule of $NADP^+$ and decarboxylates 6-phosphogluconate to the pentose, ribulose-5-phosphate.



3.5.2. Non-oxidative phase

The non-oxidative phase involves the enzymes ribulose-5-phosphate epimerase (RE), ribulose-5-phosphate isomerase (RI), transketolase (TK), and transaldolase (TA). They bring about the following reactions:

Step 1. Two molecules of ribulose-5-phosphate are converted to xylulose-5-phosphate by ribulose-5-phosphate epimerase, and a third one is converted to ribose-5-phosphate by ribulose-5-phosphate isomerase.

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